Heatmaps for Lahore Societies: Progress Report

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Motivation

- **Why?** Visualize and compare housing societies in Lahore.
- **Goals:**
  - Urban network efficiency
  - Road structure quality
  - Real estate prices
- **Good:** Interactive comparisons across societies.
- **Bad:** Missing/uneven data for some societies.

Data Sources

- **GeoJSON polygons** – Societal boundaries (from OSM or digitization).
- **Graph statistics** – Road network features (`graph_stats.csv`).
- **Real estate data** – Prices + locations from Zameen.com.
- **Challenge:** Different naming conventions.

Name Normalization

$$\text{normalize}(s) = \text{lowercase}(s) \quad \rightarrow \quad \text{remove symbols/spaces}$$

Example: `‘‘DHA_Phase-5.geojson’’` $\rightarrow$ `‘‘dha phase 5’’`

Impact

Good: Makes merging reliable.
Bad: Still fails on ambiguous names.

Network Metrics

From road graphs (`OSMnx`):

$$\text{edge\_length\_avg} = \frac{1}{N_e} \sum_e \text{length}(e)$$

$$k_{\text{avg}} = \frac{1}{N_v} \sum_v \deg(v)$$

$$\text{circuity} = \frac{\text{actual path length}}{\text{straight-line distance}}$$

Good: Quantifies road quality.

Bad: Sensitive to missing roads in OSM.

Composite Score

Convert each metric to a Z-score:

$$z_m = \frac{x_m - \mu_m}{\sigma_m}$$

If lower is better (e.g., circuity):

$$z_m = -z_m$$

Weighted average:

$$\text{Score} = \frac{\sum_m w_m \cdot z_m}{\sum_m w_m}$$

Interpretation

High score = more efficient network. **Low score** = less connectivity / more detours.

Real Estate Heatmap (DHA Example)

$$\text{AvgPrice}_p = \frac{1}{N_p} \sum_{i=1}^{N_p} \text{price}_i$$

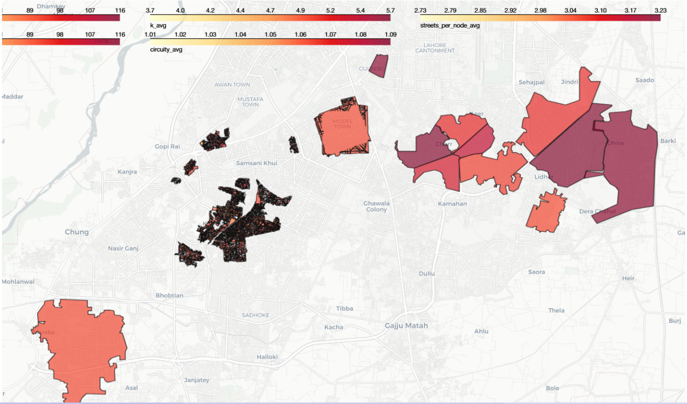
$$\text{AvgPriceCrore} = \frac{\text{AvgPricePKR}}{10^7}$$

Good: Reflects actual market value.

Bad: Biased by sparse / noisy listings.

Visualization

- **Layers (Folium):**
  - Network metrics (toggleable)
  - Composite score (default)
  - Avg price per DHA phase
- Tooltips show society name + stats.
- Colormaps highlight high vs. low performers.



Early Results & Next Steps

- **Results:**
  - Ranked societies by network efficiency.
  - DHA phases compared by avg price.
- **Next Steps:**
  - Add more factors: time-series prices, traffic, population density.
  - Incorporate additional network/urban design metrics.
  - Collect accurate society boundaries beyond DHA.
  - Move towards predictive modeling for investment hotspots.

Note & Repository

Important

This is still a **very rough prototype**.

- Many metrics remain to be added.
- Proper and complete society boundaries are needed.
- Data coverage must improve for reliable insights.

Project Repository:

https://github.com/tk1475/SPROJ